

TrES-3b

R-0393

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_250825 · created 2026-07-19T01:05:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460912.7724 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.041 +/- 0.056
Transit depth (Rp/Rs)^2	0.16 +/- 0.46 [%]
Orbital Inclination (inc)	80.14 +/- 2.9
Airmass coefficient 1 (a1)	232.28 +/- 381.87
Airmass coefficient 2 (a2)	-3.0 +/- 1.43
Scatter in the residuals of the lightcurve fit is	647.77 %
Best Comparison Star	#3 - [158, 290]
Optimal Aperture	3.0
Optimal Annulus	3.5
Transit Duration (day)	0.023 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.041 $\pm$ 0.056 vs 0.16309 (deviation 2.18 σ)
Duration fit vs published	0.023 d vs 0.0567 d (deviation 1.29 σ)
Mid-time O-C	1.9 min
Depth SNR	0.7
Residual scatter	647.77 %

Light curve

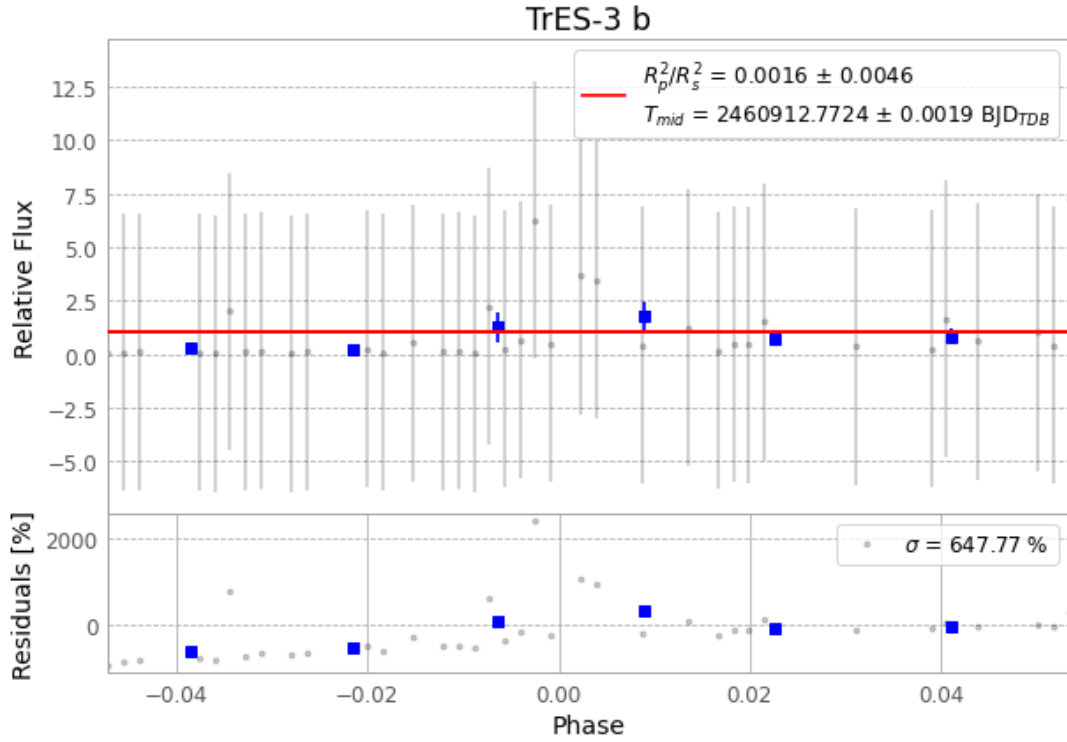


Figure 1 — FinalLightCurve_TrES-3 b_2025-08-24.png (SHA-256 35897815099a81cf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [182, 300], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1ab10491a368c161...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8350c6b

Data provenance

65 FITS frames (43 MB), TRES-3250825050016.FITS → TRES-3250825081215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-250825.log (38c7db8ef3cc...), FinalLightCurve_TrES-3 b_2025-08-24.png (35897815099a...), FinalLightCurve_TrES-3 b_2025-08-24.csv (5b6bad24007d...)

Compute resources

Wall time 170.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 01:05Z.